Cosmic Distance Ladder

Study Guide — Honors Astronomy & Astrophysics

UNL NAAP Modules 5–14 · The Glass Universe, Part III (Ch. 11–12)

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Rung 6 · Type Ia Supernovae ............... ~1 billion ly
Rung 5 · Cepheid Variables (Leavitt's Law) ~100 million ly
Rung 4 · Spectroscopic Parallax ........... ~10,000 ly
Rung 3 · Main-Sequence Fitting ........... ~50,000 ly
Rung 2 · Distance Modulus ................ ~1,000 ly
Rung 1 · Trigonometric Parallax ........... ~1,000 ly
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Exam format: 10–15 short-response questions + distance modulus calculations

Open notes · Open *The Glass Universe* · Calculator required

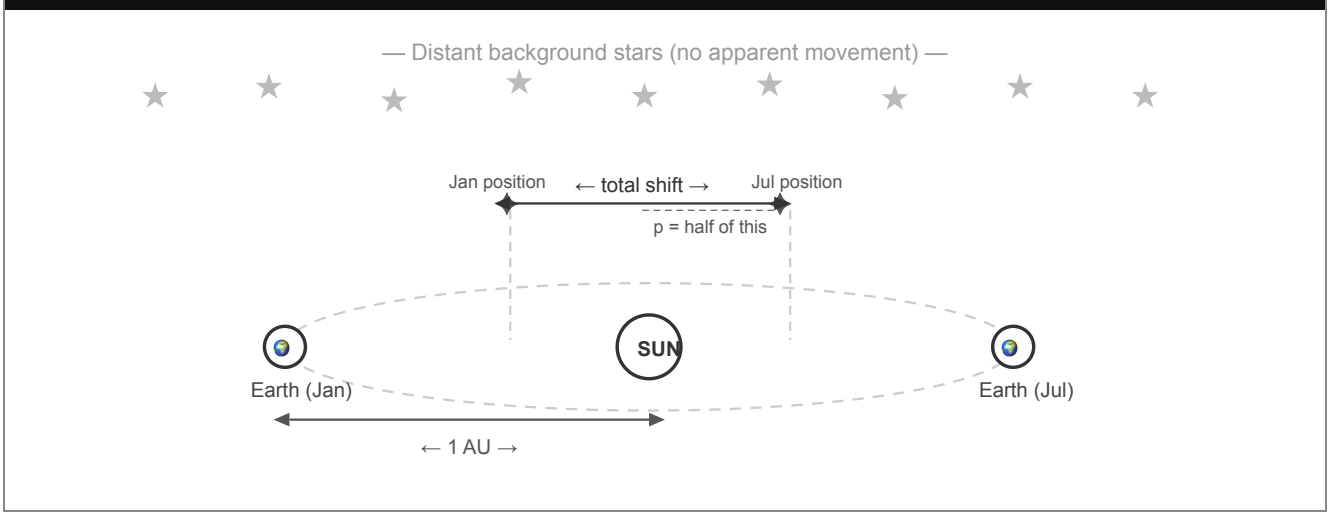
Rung 1 — Trigonometric Parallax & the Parsec

What is Trigonometric Parallax?

Parallax is the apparent shift in a nearby star's position against background stars, caused by Earth's movement around the Sun. We observe the star in January, then again in July, and measure how much it appears to shift.

- The angle of shift is the **parallax angle p**, measured in arcseconds
- Closer stars have a larger parallax angle; farther stars have a smaller angle
- The baseline is 1 AU (the Earth–Sun distance)
- Works reliably only up to about 1,000 light-years

HOW PARALLAX WORKS



What is a Parsec?

A parsec is the distance at which a star would have a parallax angle of exactly 1 arcsecond. It is the standard unit used in astronomy for stellar distances.

1 parsec = 3.26 light-years = 206,265 AU

$$d \text{ (parsecs)} = 1 \div p \text{ (arcseconds)}$$

Example: parallax angle = 0.5 arcseconds → d = 1 ÷ 0.5 = 2 parsecs = 6.52 light-years

Parallax Angle (p)	Distance (parsecs)	Distance (light-years)
1.000 arcsec	1 pc	3.26 ly
0.500 arcsec	2 pc	6.52 ly

Parallax Angle (p)	Distance (parsecs)	Distance (light-years)
0.100 arcsec	10 pc	32.6 ly
0.010 arcsec	100 pc	326 ly
< 0.003 arcsec	> 300 pc	too far — method fails

Limitation: Parallax only works reliably out to about 1,000 light-years. Beyond that, the angle is too small to measure accurately, which is why we need higher rungs of the ladder.

Rung 2 — The Distance Modulus

Apparent vs. Absolute Magnitude

Apparent Magnitude (m)

How bright a star *looks* from Earth. Depends on both the star's true brightness and its distance. Lower number = brighter.

Sun: m = −26.7 | Faintest visible to naked eye: m = +6

Absolute Magnitude (M)

How bright a star *truly is* — its intrinsic luminosity. Defined as what its apparent magnitude would be if it were placed exactly 10 parsecs away.

Sun: M = +4.83 | Rigel: M = −7.0

The Formulas

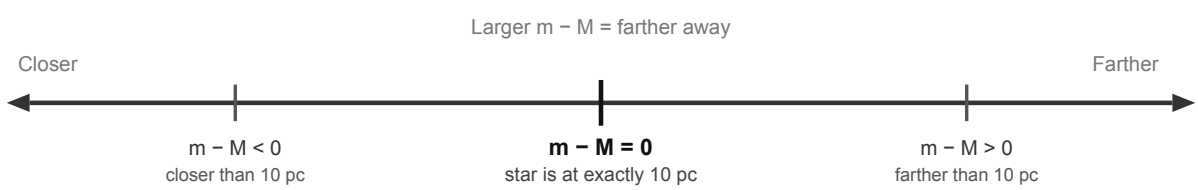
$$m - M = 5 \times \log_{10}(d) - 5$$

m = apparent magnitude | M = absolute magnitude | d = distance in parsecs

$$d = 10 \wedge ((m - M + 5) \div 5)$$

Use this form when you need to solve for distance

UNDERSTANDING THE DISTANCE MODULUS (M − M)



Distance Modulus Reference Table

m − M	Distance (parsecs)	Distance (light-years)	Where this puts you
0	10 pc	32.6 ly	Reference point (definition of M)
5	100 pc	326 ly	Local stellar neighborhood
10	1,000 pc	3,260 ly	Within the Milky Way
15	10,000 pc	32,600 ly	Milky Way disk

m − M	Distance (parsecs)	Distance (light-years)	Where this puts you
20	100,000 pc	326,000 ly	Magellanic Clouds
25	1,000,000 pc (1 Mpc)	3.26 million ly	Andromeda Galaxy region

Worked Examples

Example 1 — Find distance

A star has m = +8.5 and M = +3.5. Find its distance.

m − M = 8.5 − 3.5 = **5.0**

d = 10^((5.0 + 5) ÷ 5) = 10^2 = **100 pc** = 326 light-years

Example 2 — Negative M (bright star)

A Cepheid has m = +12, M = −3. Find its distance.

m − M = 12 − (−3) = 12 + 3 = **15**

d = 10^((15 + 5) ÷ 5) = 10^4 = **10,000 pc** = 32,600 light-years

Example 3 — Find apparent magnitude

Star is at 500 pc, M = +2. What does it look like from Earth?

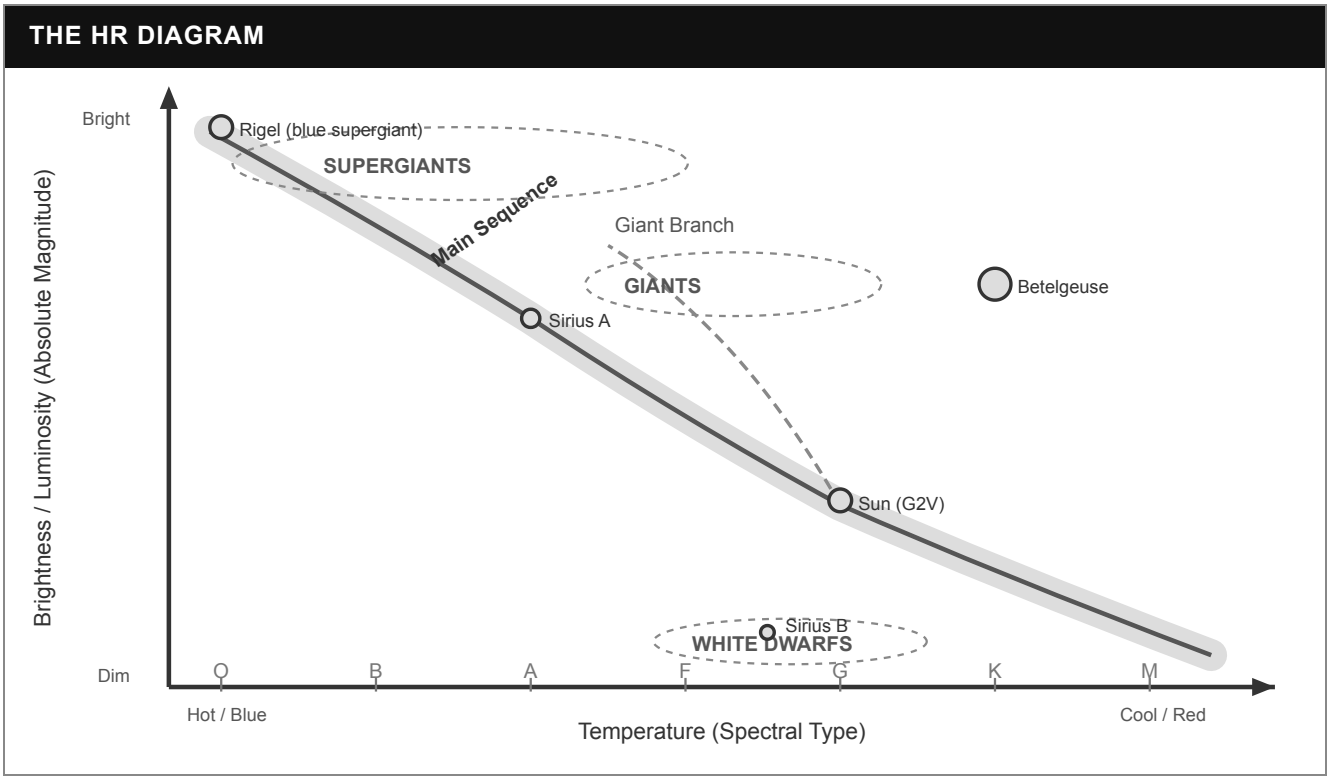
m − M = 5 × log(500) − 5 = 5 × 2.699 − 5 = 13.495 − 5 = **8.495**

m = 2 + 8.495 = **+10.5**

Rungs 2–3 — HR Diagrams, Main-Sequence Fitting & Cluster Ages

The Hertzsprung-Russell (HR) Diagram

A graph that plots stars by temperature (x-axis, hot on the left to cool on the right) against luminosity or absolute magnitude (y-axis, dim at the bottom to bright at the top). Most stars fall along a diagonal band called the main sequence.



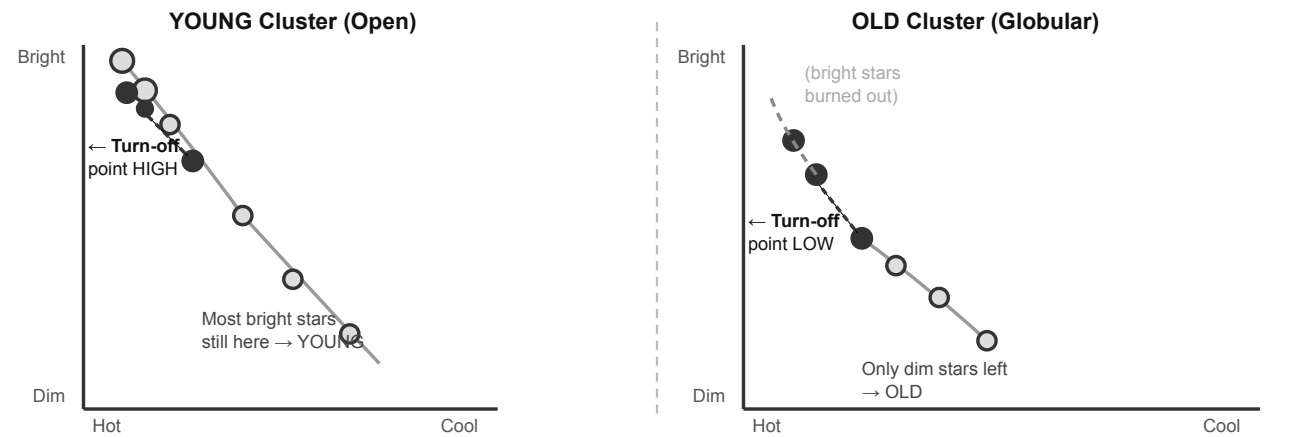
Main-Sequence Fitting

To find the distance to a star cluster, plot the cluster's stars on an HR diagram using apparent magnitude on the y-axis. Then compare this to the standard main sequence plotted using absolute magnitude. The vertical shift needed to line them up equals the distance modulus (m − M), which you can then use to calculate the distance.

The Main-Sequence Turn-Off Point

The turn-off point is where a cluster's stars begin to leave the main sequence and move toward the giant branch. Massive, luminous stars consume their fuel fastest and leave the main sequence first. The location of the turn-off point tells us the age of the cluster.

TURN-OFF POINT — YOUNG VS. OLD CLUSTER



Open vs. Globular Clusters

	Open Clusters	Globular Clusters
Size	Hundreds–thousands of stars	Tens of thousands–millions of stars
Location	Galactic disk/spiral arms	Galactic halo (spherical distribution)
Age	Young — millions to a few billion years	Old — 10–13 billion years
Turn-off point	HIGH on HR diagram (bright stars present)	LOW on HR diagram (only dim stars remain)
Age determination	✔ Easier — clear, sharp turn-off	Harder — complex HR diagram
Examples	Pleiades, Hyades	M13, Omega Centauri

Rung 4 — Spectroscopic Parallax

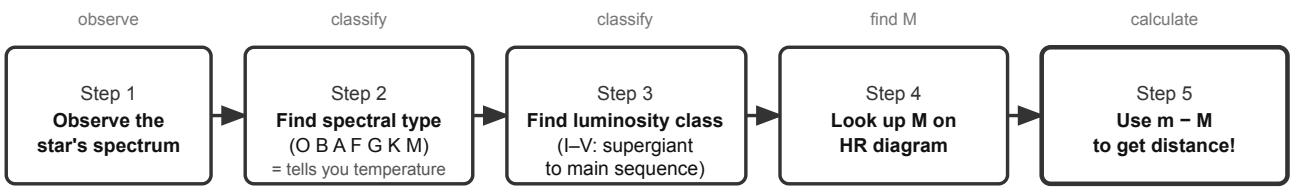
Important: The term "spectroscopic parallax" is misleading. No parallax angle is measured. The method uses the star's spectrum to determine its absolute magnitude, then applies the distance modulus. It is called "parallax" only because it serves the same purpose — finding distance.

Why We Need It

For a single star not in a cluster, main-sequence fitting doesn't work. If the star is too far away, the parallax angle is too small to measure. Spectroscopic parallax solves this by reading the star's true brightness from its spectrum.

How It Works

SPECTROSCOPIC PARALLAX — STEP BY STEP



Luminosity Classes

Class	Type of Star	Example	Absolute Magnitude (approx.)
I (Ia/Ib)	Supergiant — very large and bright	Betelgeuse	−5 to −7
II	Bright Giant	Canopus	−2 to −4
III	Giant	Arcturus	0 to −2
IV	Subgiant	Procyon	+2 to +3
V	Main Sequence (Dwarf) — normal star	Sun (G2V)	+4.83

Cecilia Payne & Why Her Work Matters

Cecilia Payne-Gaposchkin (1900–1979)

First Harvard Astronomy PhD · Thesis: Stellar Atmospheres (1925)

- Proved that spectral types (O B A F G K M) are caused by **temperature differences**, not different chemical compositions
- Discovered that stars are made primarily of **hydrogen and helium** — a conclusion that contradicted prevailing assumptions
- Henry Norris Russell pressured her to soften her conclusions; he later confirmed she was right and received much of the credit
- Her work is the direct foundation of spectroscopic parallax: since spectral type = temperature, and temperature determines position on the HR diagram, we can read off absolute magnitude M from a star's spectrum alone

Her thesis was later described as "the most brilliant PhD thesis ever written in astronomy."

Rung 5 — Leavitt's Law & Cepheid Variables

What Are Cepheid Variables?

Cepheid variables are pulsating giant stars that rhythmically brighten and dim on a regular, repeating schedule. They physically expand and contract. The time between peak brightnesses is the period, which ranges from about 1 to 100 days.

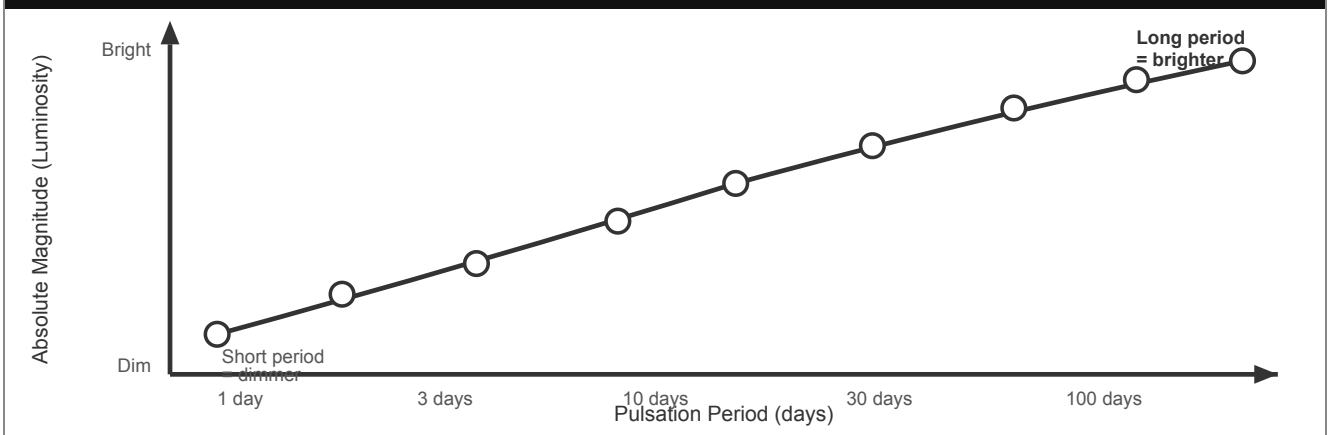
Henrietta Swan Leavitt (1868–1921)

Harvard "Computer" — analyzed glass photographic plates, paid 25 cents per hour

- Studied photographic plates of the Small Magellanic Cloud (SMC), a nearby dwarf galaxy
- Because all SMC stars are at approximately the same distance from Earth, any difference in apparent brightness reflects a real difference in intrinsic luminosity
- Discovered 25 Cepheid variables and found a clear pattern: **longer period = greater luminosity**
- Published in 1912. Discovered over 2,400 variable stars total. Nominated for a Nobel Prize (awarded posthumously — she had already died)

Leavitt's Law (Period-Luminosity Relation): The longer a Cepheid's pulsation period, the greater its intrinsic luminosity (absolute magnitude M). Measure the period, look up M, measure apparent magnitude m, and apply the distance modulus to find distance.

LEAVITT'S LAW — PERIOD VS. LUMINOSITY



Calibrating Leavitt's Law

To turn the period-luminosity relationship into actual distances (not just relative distances), astronomers had to find nearby Cepheids whose distances were already known from trigonometric parallax. Using those known distances plus observed apparent magnitudes, they calculated actual

absolute magnitudes M for each period. This calibrated the law so that any Cepheid anywhere in the universe can yield a real distance. This is an example of a lower ladder rung (parallax) calibrating a higher one (Cepheids).

The Great Debate

In 1920, astronomers did not know the true scale of the universe. The central question was whether spiral "nebulae" — fuzzy patches in the sky — were gas clouds inside our own Milky Way, or separate galaxies far beyond it.

Harlow Shapley — "Nebulae are inside the Milky Way"

- Argued the Milky Way was enormous and essentially the whole universe
- Was correct that the Sun is not at the center of the Milky Way
- Was wrong about the overall scale of the universe

Heber Curtis — "They are separate galaxies"

- Argued spiral nebulae are distant "island universes" — galaxies in their own right
- Pointed to dust lanes and nova brightness as evidence
- Was correct — confirmed by Hubble in 1923–1924

Edwin Hubble — How He Settled It (1923–1924)

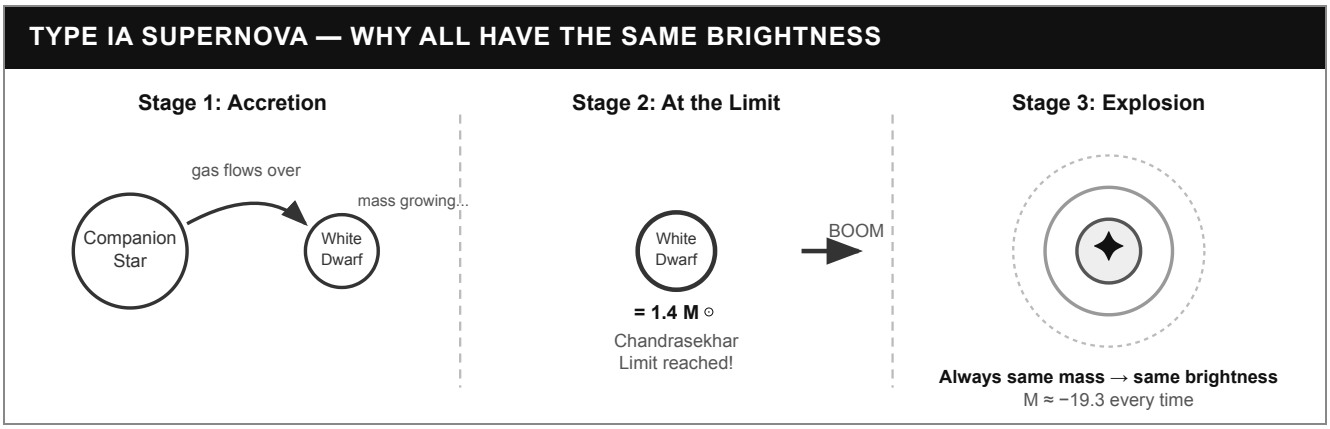
Mount Wilson Observatory · 100-inch Hooker Telescope

- Identified Cepheid variable stars in the Andromeda "Nebula" (M31)
- Applied Leavitt's Law to calculate the distance: approximately 900,000 light-years (modern value: 2.5 million light-years)
- This placed Andromeda far beyond any possible boundary of the Milky Way, proving it is a separate, independent galaxy
- Curtis was right. The universe contains billions of galaxies.
- Hubble's discovery was only possible because of Leavitt's period-luminosity law

Rung 6 — Type Ia Supernovae

What Is a Type Ia Supernova?

A Type Ia supernova occurs in a binary star system where a white dwarf gradually accretes (pulls in) mass from its companion star. When the white dwarf reaches **1.4 solar masses — the Chandrasekhar limit** — it undergoes a runaway thermonuclear explosion and is completely destroyed.



Why Type Ia Supernovae Are Useful for Distance

Because the Chandrasekhar limit (1.4 solar masses) is set by quantum mechanics and is the same everywhere in the universe, every Type Ia supernova explodes with the same amount of fuel and releases the same peak energy. This makes them **standard candles** — objects with a known, consistent absolute magnitude ($M \approx -19.3$). Observe the apparent magnitude m, apply the distance modulus, and you can find distances to galaxies billions of light-years away.

Type II supernovae (core collapse of massive stars) have variable brightness and cannot be used for distance measurement.

Master Reference Sheet

Core Formulas

$$d = 1 \div p$$

parallax: d (parsecs), p (arcseconds)

$$m - M = 5 \log(d) - 5$$

distance modulus

Key People

Person	Contribution
Leavitt	Period-luminosity law (1912) — allows distance to any galaxy with a Cepheid

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d = 10^((m−M+5)÷5)
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→ distance in parsecs

Key Numbers

1 parsec	3.26 light-years
m−M = 5	100 pc away
m−M = 10	1,000 pc away
m−M = 15	10,000 pc away
Chandrasekhar limit	1.4 M ☉
Type Ia peak M	≈ −19.3

Person	Contribution
Payne	Spectral type = temperature; stars = H + He — foundation of spectroscopic parallax
Hubble	Found Cepheids in Andromeda (1923–24) — settled Great Debate
Shapley	Great Debate: nebulae inside Milky Way — WRONG about scale
Curtis	Great Debate: nebulae are separate galaxies — CORRECT

Method Ranges

Method	Max Range
Trig. Parallax	~1,000 ly
Main-Seq Fitting	~50,000 ly
Spectroscopic Parallax	~10,000 ly
Cepheids / Leavitt's Law	~100 million ly
Type Ia Supernovae	>1 billion ly

Things to Remember for the Exam

- "Spectroscopic parallax" does not involve measuring a parallax angle — it uses spectral analysis
- A turn-off point LOW on the HR diagram means the cluster is OLD (bright stars burned out)
- Open clusters are easier to determine ages for than globular clusters
- Larger m − M means the star is farther away
- Leavitt's Law must be calibrated using parallax distances to nearby Cepheids
- Payne proved spectral type = temperature, not chemical composition
- Type Ia supernovae always explode at 1.4 M ☉ — so they always have the same peak brightness
- Hubble used Leavitt's Law to prove the Great Debate was won by Curtis

Good luck. · Calculator + The Glass Universe + Binder of Notes